

GOVT 6019: Rolling Dice — Solutions

Week 5 in-class activity · Fall 2026

Part 1. Before you roll

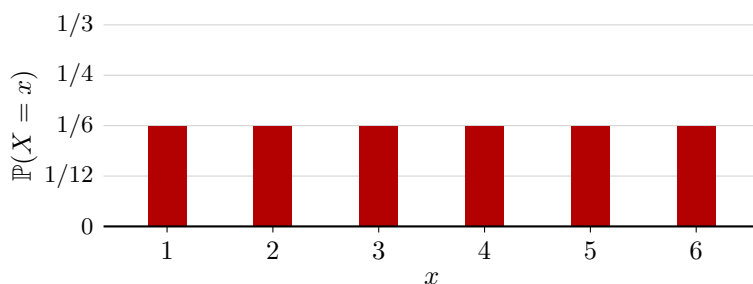
Let X be the result of one roll of a fair die. Each face has probability $1/6$.

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5$$

$$\mathbb{E}[X^2] = \frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6}, \quad \mathbb{V}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12} \approx 2.92$$

The standard deviation is $\sigma = \sqrt{35/12} \approx 1.71$.

Figure A. The probability mass function of X : six bars, each of height $1/6$.



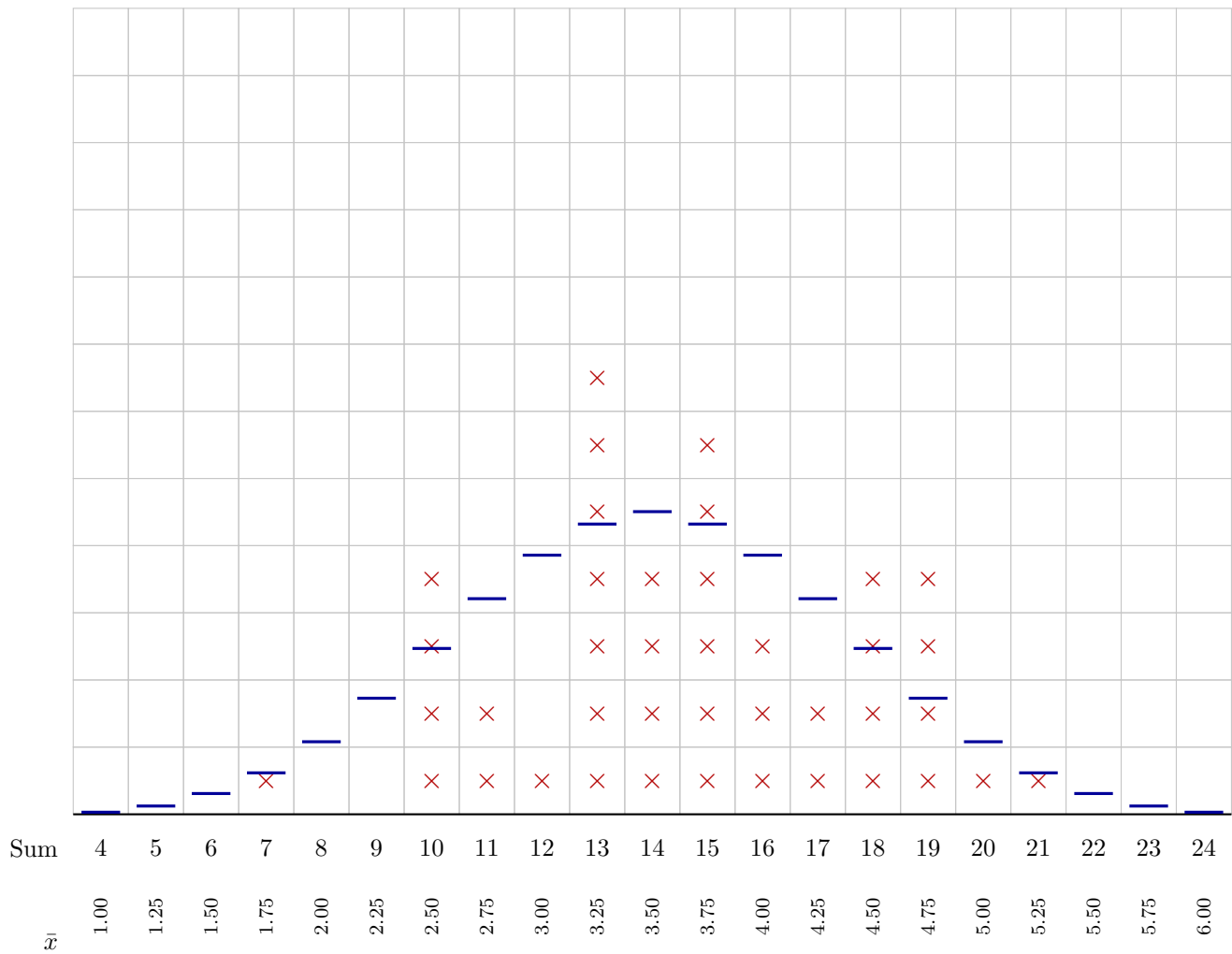
Part 2. Roll

Forty trials of four rolls each. Each cell holds the sum of one trial.

Log of sums

¹ 10	² 13	³ 18	⁴ 15	⁵ 10	⁶ 15	⁷ 13	⁸ 11	⁹ 13	¹⁰ 10
¹¹ 15	¹² 19	¹³ 16	¹⁴ 14	¹⁵ 19	¹⁶ 13	¹⁷ 18	¹⁸ 18	¹⁹ 13	²⁰ 11
²¹ 15	²² 17	²³ 14	²⁴ 14	²⁵ 16	²⁶ 15	²⁷ 17	²⁸ 19	²⁹ 10	³⁰ 19
³¹ 14	³² 7	³³ 13	³⁴ 18	³⁵ 13	³⁶ 16	³⁷ 21	³⁸ 12	³⁹ 15	⁴⁰ 20

Figure B. The means of the 40 trials. Each $\times$ is one trial. The blue line in each column marks how high the stack is expected to reach in 40 trials, $40 \times \mathbb{P}(\text{sum})$.



	These 40 trials	Theory
Mean of the $\bar{x}$'s	3.68	$\mu = 3.50$
Standard deviation of the $\bar{x}$'s	0.81	$\sigma/\sqrt{4} = 0.85$
Smallest and largest sum	7 and 21	4 and 24 possible

Part 3. Questions

Recall the premise of the Central Limit Theorem (CLT): the sampling distribution of the sample means (i.e., the distribution of $\bar{x}$'s) will follow a Normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$, regardless of the distribution of the population. Because we know the distribution of the values of the die and this distribution is symmetric, we can see how the CLT works with a small sample size ($n = 4$).

We will now compare the results of rolling one die versus the activity you performed (the mean of four rolls of a die).

1. Comment on the difference in shape between Figure A and Figure B.

Answer. Figure A is flat. Every value from 1 to 6 has the same probability, $1/6$. In Figure B the $\times$'s pile up in the middle and thin out toward both ends. 25 of the 40 $\times$'s sit between sums of 11 and 17, which is $\bar{x}$ between 2.75 and 4.25. The smallest sum in these 40 trials was 7 and the largest was 21. No trial came near 4 or 24.

The extremes are rare because few combinations of four rolls produce them. A sum of 4 needs four 1s, which is 1 of the $6^4 = 1,296$ equally likely outcomes. A sum of 14 can be made 146 ways. A single roll of 1 is as likely as a roll of 3. A *mean* of 1 needs every roll to be low at once.

2. Do you believe that the Central Limit Theorem is working with regards to the shape of Figure B?

Answer. Yes. Figure B is already roughly bell-shaped at $n = 4$. The $\times$'s pile up in the middle, and the counts fall off on both sides. The mean of the 40 $\bar{x}$'s was 3.68, close to $\mu = 3.5$. It is not exactly Normal. $\bar{x}$ can take only 21 values, and it can never fall below 1 or above 6. A Normal distribution is continuous and unbounded.

Forty trials is a small number, so the $\times$'s are lumpy. Some stacks rise above their blue line and some fall short. Sum 14, for example, is the most likely sum but got only 4 trials here. Those gaps and bumps are noise from having only 40 trials. They are not features of the sampling distribution. The blue lines show its true shape, and they trace a smooth bell. The CLT works this fast here because the die's distribution is symmetric. A skewed population would need a larger n .

3. If the population has a standard deviation of σ and the sampling distribution of the sample means has a standard deviation of $\sigma/\sqrt{n}$ (according to the CLT), which one of these two values will be lower if $n > 1$? In other words, which of these two distributions do you expect to have lower variability: the population distribution or the sampling distribution of the sample means?

Answer. $\sigma/\sqrt{n}$ is lower. When $n > 1$, $\sqrt{n} > 1$, so dividing σ by $\sqrt{n}$ gives a smaller number. The sampling distribution of the sample means has lower variability than the population.

For the die, $\sigma = 1.71$. With $n = 4$ the standard deviation of $\bar{x}$ is $1.71/\sqrt{4} = 1.71/2 = 0.85$, half of σ . In these 40 trials the standard deviation of the $\bar{x}$'s was 0.81. The figures show the same comparison. In Figure A the probability is spread evenly from 1 to 6. In Figure B 33 of the 40 $\times$'s lie between 2.5 and 4.5.